

國立中央大學數學系 114 學年度專題海報競賽

Calibration and Predictive Performance of Continuous-Time and Discrete-Time Stochastic Volatility Models in Option Pricing

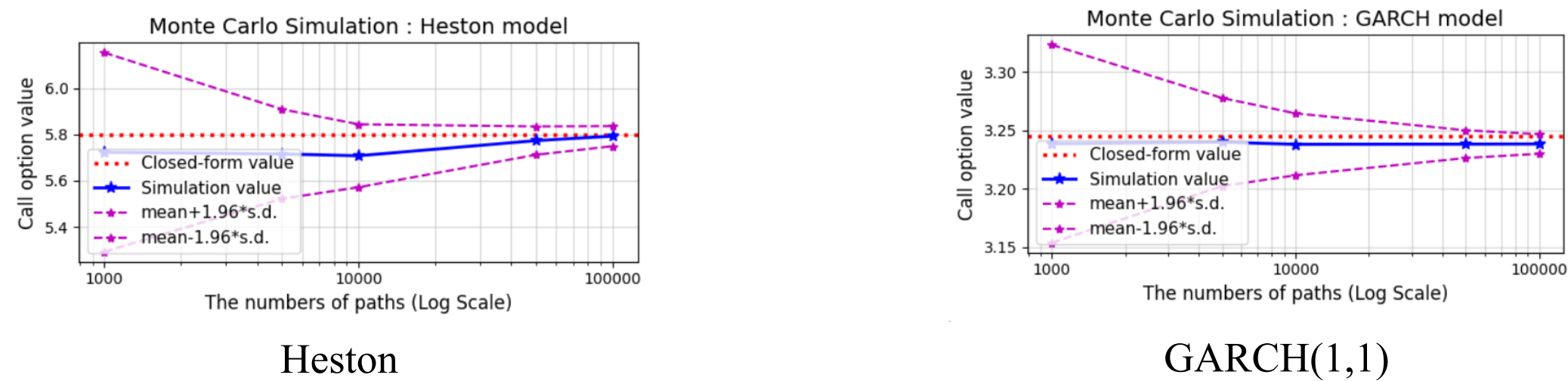
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Introduction

This project evaluates how time-structural differences affect option pricing by comparing the continuous-time Heston model against the discrete-time Heston-Nandi GARCH(1,1) model. We investigate how these two distinct frameworks adapt to market shocks under severe stress.Dictate their calibration resilience and out-of-sample predictive performance.

Closed-Form Verification

Using Monte Carlo simulation through random sampling to construct 95% confidence interval.



The above charts show that both models converge to the closed-form option prices (red line) within the 95% confidence intervals.

Heston Model Dynamics & Characteristics Function

Stock dynamics: $dS(t) = \mu Sdt + \sqrt{v(t)}SdW_1(t)$
Volatility dynamics: $dv(t) = \kappa(\theta - v(t))dt + \sigma\sqrt{v(t)}dW_2(t)$
Where $W_1(t), W_2(t)$ are Wiener processes, and $Cov[W_1(t), W_2(t)] = \rho dt, \rho \in [-1, 1]$
The characteristics function is:
 $f(x, v, \tau, \phi) = \exp(C(\tau, \phi) + D(\tau, \phi)v + i\phi x), x = \ln S$
 $\tau = T - t$ is the time to maturity.

Heston FFT Closed-Form Derivation

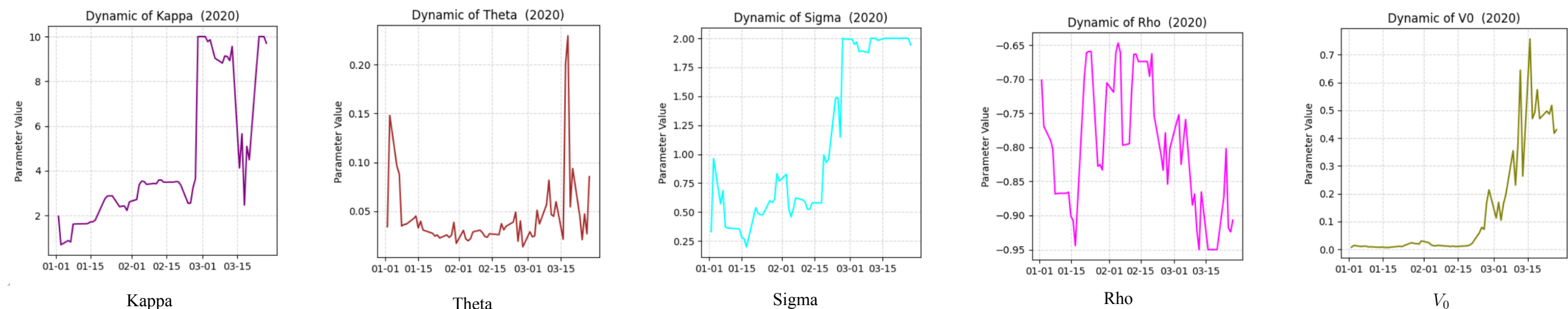
$$C(t, K) = e^{-r\tau} \int_k^\infty (e^{s_T} - e^k) Q(s_T) ds_T \quad , \text{ where } s_T = \ln S_T, k = \ln K$$

Since $Q(s_T)$ is unknown, thus we transit to the Fourier domain via $f_\tau(\omega) = E[e^{i\omega s_T}]$
$$\psi_\tau(\omega) = \mathcal{F}(e^{\alpha k} C(t, k)) = \int_{-\infty}^\infty e^{i\omega k} e^{\alpha k} C(t, k) dk = \frac{e^{-r\tau} f_\tau(\omega - i(\alpha + 1))}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega}, \alpha > 0$$

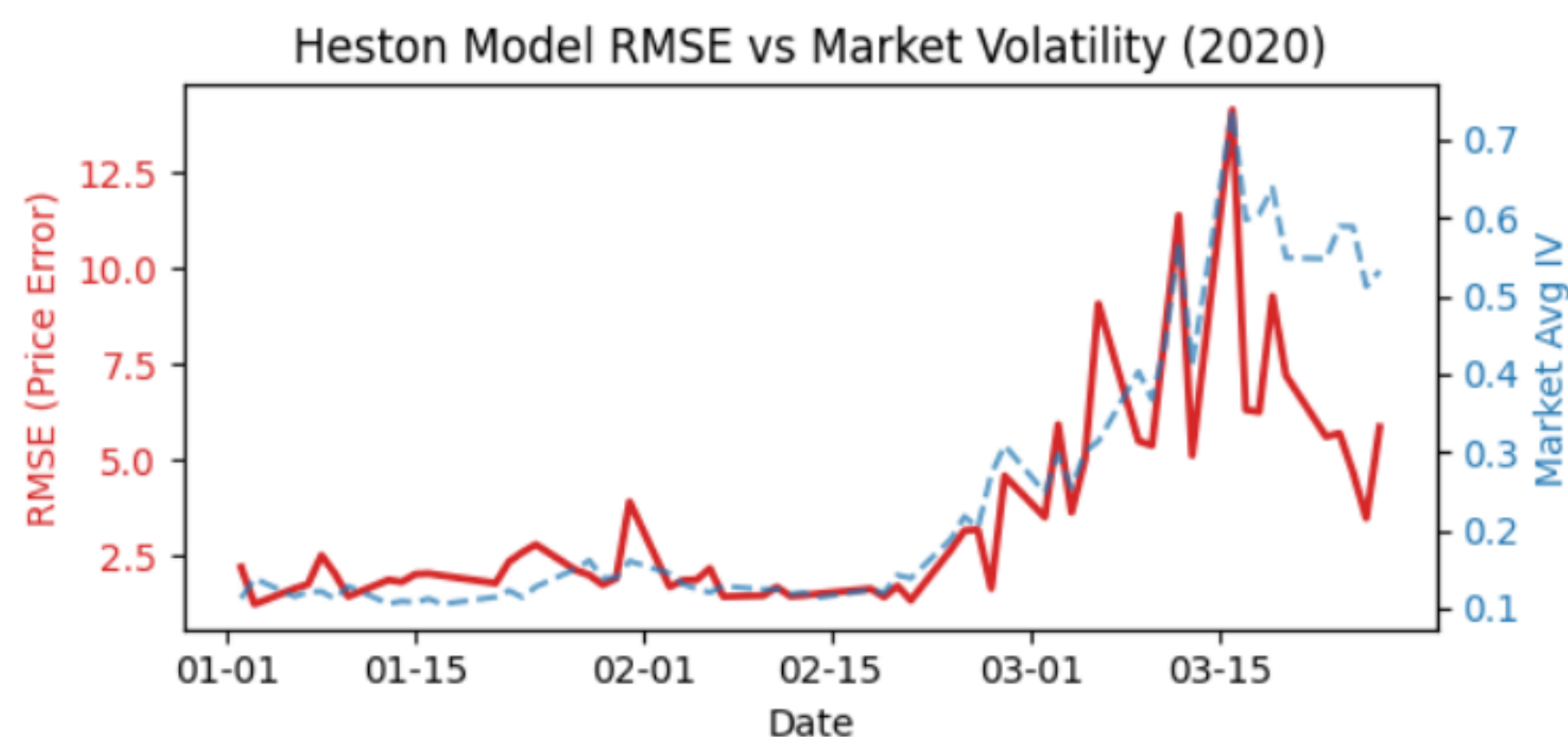
$$C(t, k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^\infty e^{-i\omega k} \psi_\tau(\omega) d\omega = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^\infty e^{-i\omega k} \frac{e^{-r\tau} f_\tau(\omega - i(\alpha + 1))}{\alpha^2 + \alpha - \omega^2 + i(2\alpha + 1)\omega} d\omega$$

Heston Calibration Results

Define parameter $\Theta^H = (\kappa, \theta, \sigma, \rho, v_0)$, $\hat{\Theta}_t^H = \arg \min_{\Theta^H} \|\mathbf{C}_t^{\text{model}}(\Theta^H) - \mathbf{C}_t^{\text{market}}\|^2$
Parameters are calibrated by minimizing pricing errors between model and market prices.
Dynamics of parameters:

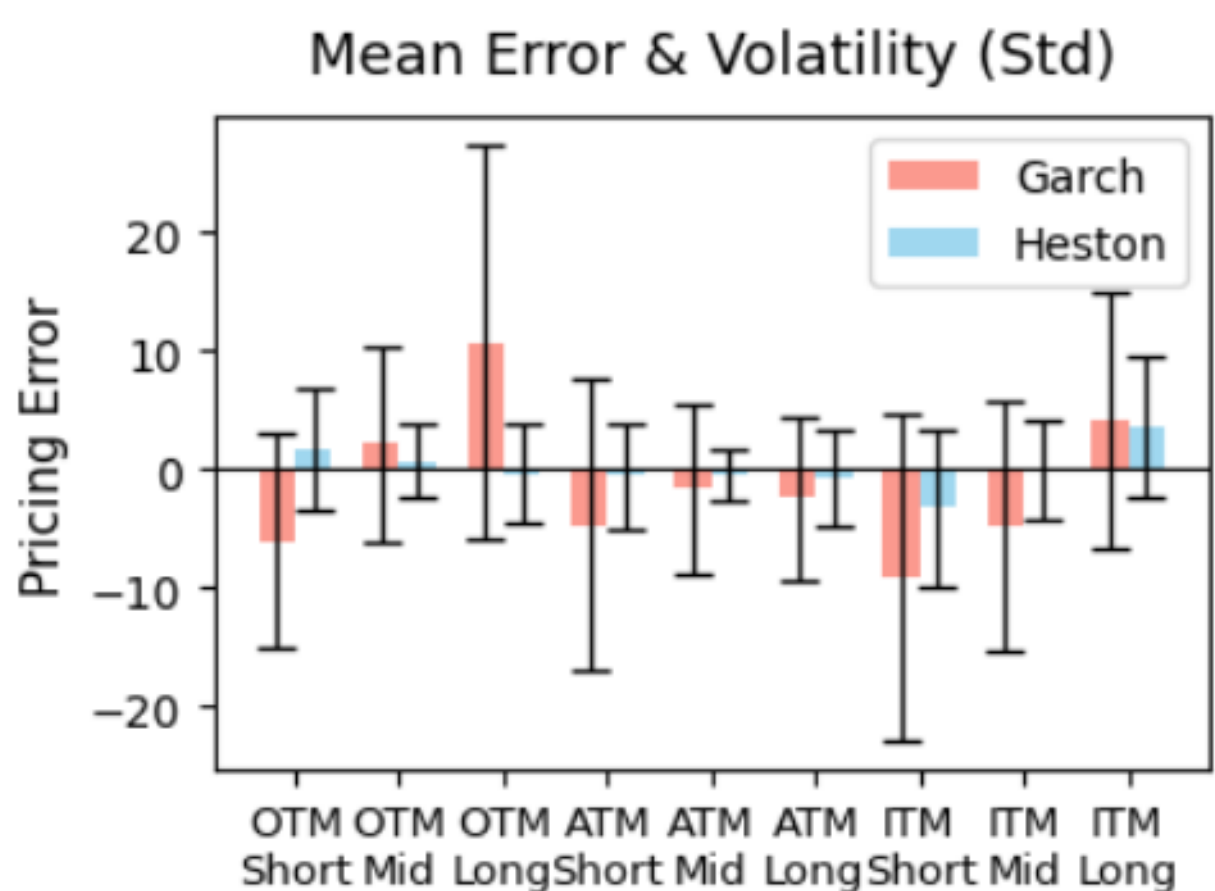


The sharp spike in v_0 and σ during March 2020 reflects the sudden explosion of market uncertainty and volatility caused by the COVID-19 crisis.



High RMSE values coincide with periods of elevated market volatility, with March 2020 representing the peak of the COVID-19 market crisis.

In Sample Performance Comparison



- Heston model demonstrates stable pricing performance across different maturities. Because its continuous diffusion path can better capture the market volatility.
- GARCH model performs worse for long-maturity OTM options, less stable outside ATM options, because its variance updates in discrete daily intervals.

GARCH(1,1) Dynamics & Characteristics Function

$$\ln S(t) = \ln S(t - \Delta) + r - \frac{1}{2}h(t) + \sqrt{h(t)}z^*(t)$$

$$h(t) = \omega + \beta_1 h(t - \Delta) + \alpha_1 (z^*(t - \Delta) - \gamma_1^* \sqrt{h(t - \Delta)})^2$$
 where $\gamma_1^* = \gamma_1 + \lambda + \frac{1}{2}, z^*(t) \sim N(0, 1)$
$$A(t; T, \phi) = A(t + \Delta; T, \phi) + \phi r + B_1(t + \Delta; T, \phi)\omega - \frac{1}{2} \ln(1 - 2\alpha_1 B_1(t + \Delta; T, \phi))$$

$$B_1(t; T, \phi) = \phi(\lambda + \gamma_1) - \frac{1}{2}\gamma_1^2 + \beta_1 B_1(t + \Delta; T, \phi) + \frac{1/2(\phi - \gamma_1)^2}{1 - 2\alpha_1 B_1(t + \Delta; T, \phi)}$$

$$f(\phi) = S(t)^\phi \exp(A(t; T, \phi) + B_1(t; T, \phi)h(t + \Delta))$$

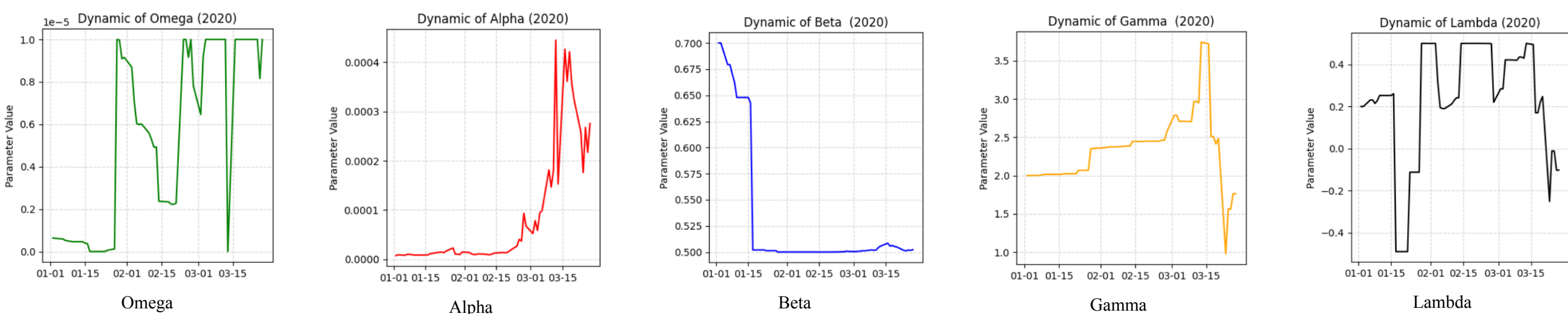
GARCH(1,1) Closed-Form Derivation

Let $x = \ln(S(T))$, and $p(x)$ be the probability density
Define the moment generating function $f(\phi) = E_t[e^{\phi x}], p^*(x) = \frac{e^x p(x)}{f(1)}$
then the mgf of $p^*(x)$ is $\int_{-\infty}^\infty e^{\phi x} p^*(x) dx = \frac{1}{f(1)} \int_{-\infty}^\infty e^{(\phi+1)x} p(x) dx = \frac{f(\phi+1)}{f(1)}$
$$C = e^{-r(T-t)} E[\max(e^x - K, 0)] = f(1) \int_{\ln(K)}^\infty p^*(x) dx - K \int_{\ln(K)}^\infty p(x) dx$$

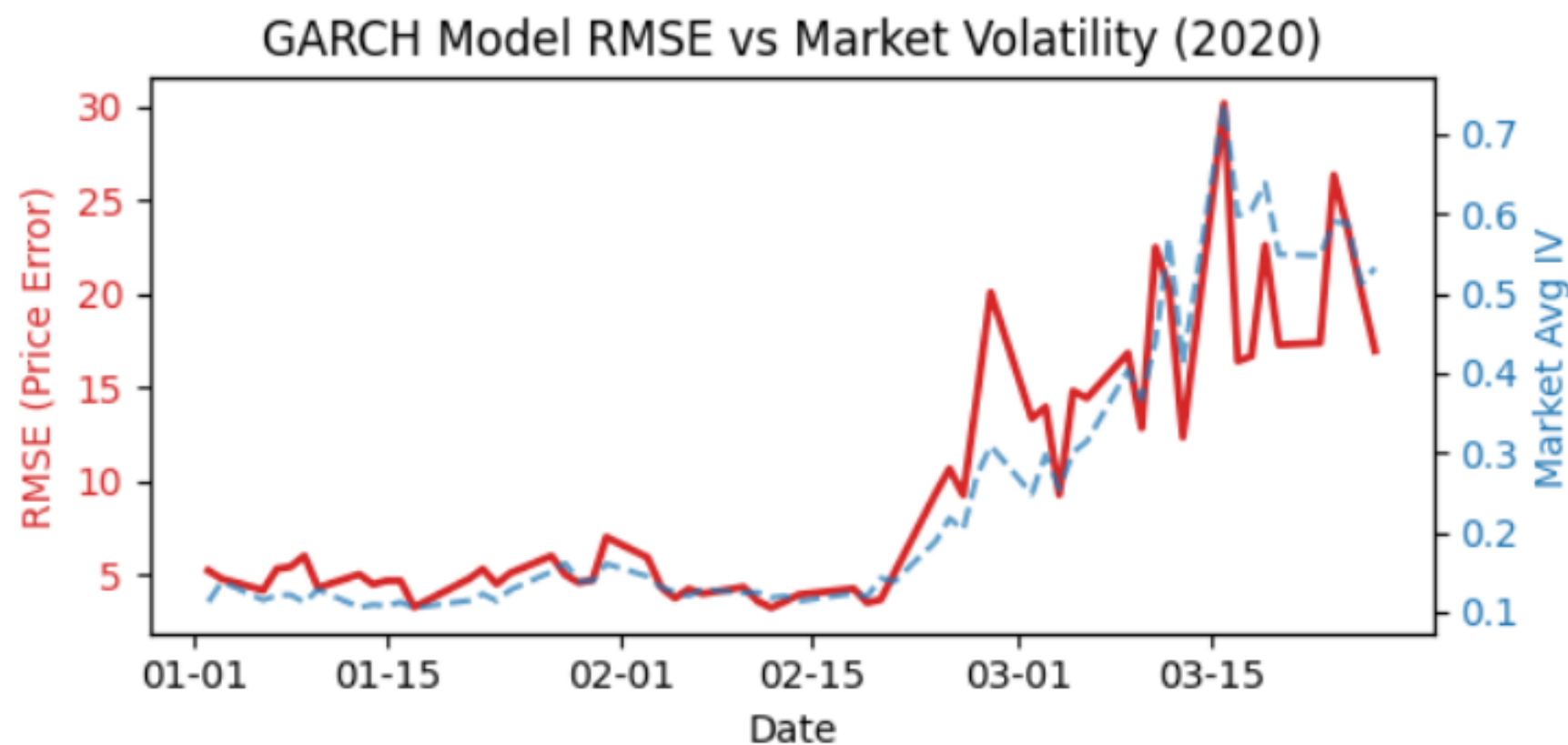
$$= \frac{1}{2} S(t) + \frac{e^{-r(T-t)}}{\pi} \int_0^\infty \text{Re} \left[\frac{K^{-i\phi} f(i\phi + 1)}{i\phi} \right] d\phi - K e^{-r(T-t)} \left(\frac{1}{2} + \frac{1}{\pi} \int_0^\infty \text{Re} \left[\frac{K^{-i\phi} f(i\phi)}{i\phi} \right] d\phi \right)$$

GARCH(1,1) Calibration Results

Define $\Theta^G = (\omega, \alpha, \beta, \gamma, \lambda)$, $\hat{\Theta}_t^G = \arg \min_{\Theta^G} \|\mathbf{C}_t^{\text{model}}(\Theta^G) - \mathbf{C}_t^{\text{market}}\|^2$
Parameters are calibrated by minimizing pricing errors between model and market option prices.
Dynamic of parameters:



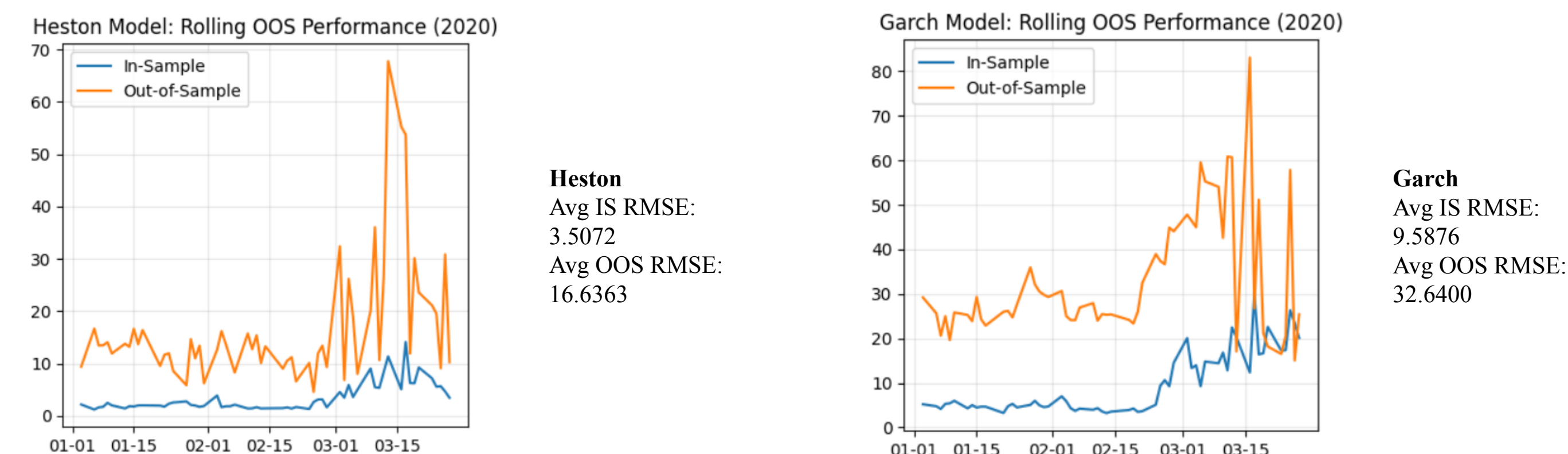
The significant increase in α and γ during March 2020 indicates heightened sensitivity to market shocks and intensified volatility clustering during the COVID-19 crisis.



The chart shows that RMSE spikes coincide with periods of extreme market volatility during the COVID-19 crisis.

Out of Sample Performance

Calculate $RMSE = \frac{1}{\sqrt{N_t}} \|\mathbf{C}_t^{\text{model}}(\Theta_{t-1}) - \mathbf{C}_t^{\text{market}}\|$ to measure the models performance.



Both models exhibit significant RMSE spikes during the COVID-19 crisis (March 2020). However, the Heston model maintains relatively lower and more stable out-of-sample errors due to the mean reversion system of volatility.

Conclusion

The empirical results show that the Heston model provides more stable and consistent option pricing performance during the COVID-19 period. As a continuous-time stochastic volatility model, Heston captures rapid changes in market volatility more effectively than the discrete-time GARCH(1,1) model. Consequently, Heston achieves lower pricing errors and better approximates observed market option prices under extreme market conditions.

References

- [1] Steven L. Heston(1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options
- [2] Steven L. Heston and Saikat Nandi (2000). A Closed-Form GARCH Option Valuation Model